Dividing with Unit Fractions

Share a piece, or count how many pieces fit.

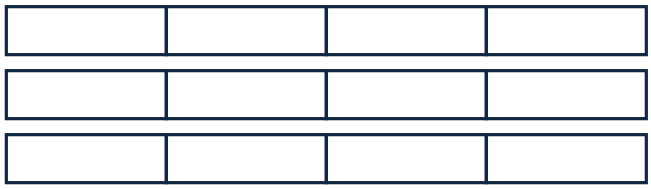
1/3 ÷ 2: share one third between 2 people



Cut every third into 2.
Now the whole has 6 pieces.
Each person gets 1 of them.

$$\frac{1}{3} \div 2 = \frac{1}{6}$$

3 ÷ 1/4: how many fourths fit in 3 wholes?



Each whole has 4 fourths.
3 wholes have 3 × 4 = 12 fourths.

$$3 \div \frac{1}{4} = 12$$

Check with times

$$\frac{1}{6} \times 2 = \frac{2}{6} = \frac{1}{3}$$

$$12 \times \frac{1}{4} = \frac{12}{4} = 3$$

Your turn!

$$\frac{1}{2} \div 4 = \frac{\boxed{1}}{\boxed{8}}$$

For teachers

Goal: divide a unit fraction by a whole number and a whole number by a unit fraction (5.NF.7). Say “Share the piece: cut it smaller” and “How many pieces fit?” No flip-and-multiply rule.

Dividing with Unit Fractions: Practice

Share a piece: it gets smaller. Pieces in wholes: count them.

Part 1 Share a piece

Every piece is cut. What part of the whole is one small piece?

1) 

Share the shaded part among 2.

$\frac{1}{4} \div 2 = \frac{1}{8}$

2) 

Share the shaded part among 3.

$\frac{1}{2} \div 3 = \frac{1}{6}$

3) 

Share the shaded part among 3.

$\frac{1}{3} \div 3 = \frac{1}{9}$

4) 

Share the shaded part among 2.

$\frac{1}{5} \div 2 = \frac{1}{10}$

5) 

Share the shaded part among 5.

$\frac{1}{2} \div 5 = \frac{1}{10}$

Part 2 Share a piece

No help this time.

6) $\frac{1}{4} \div 3 = \frac{1}{12}$

7) $\frac{1}{2} \div 6 = \frac{1}{12}$

8) $\frac{1}{5} \div 4 = \frac{1}{20}$

9) $\frac{1}{3} \div 5 = \frac{1}{15}$

10) $\frac{1}{6} \div 2 = \frac{1}{12}$

11) $\frac{1}{8} \div 2 = \frac{1}{16}$

Dividing with Unit Fractions: Practice (continued)

Part 2 Share a piece (continued)

12) $\frac{1}{2} \div 7 = \frac{1}{14}$

13) $\frac{1}{4} \div 5 = \frac{1}{20}$

14) $\frac{1}{3} \div 6 = \frac{1}{18}$

15) $\frac{1}{10} \div 3 = \frac{1}{30}$

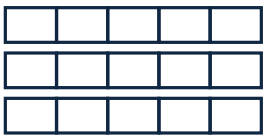
16) $\frac{1}{5} \div 5 = \frac{1}{25}$

17) $\frac{1}{6} \div 4 = \frac{1}{24}$

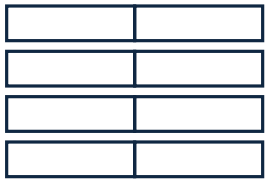
Part 3 How many pieces fit? Count all the pieces.

18) 

$2 \div \frac{1}{3} = 6$

19) 

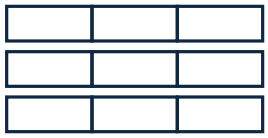
$3 \div \frac{1}{5} = 15$

20) 

$4 \div \frac{1}{2} = 8$

21) 

$2 \div \frac{1}{6} = 12$

22) 

$3 \div \frac{1}{3} = 9$

Dividing with Unit Fractions: Practice (continued)

Part 4 How many pieces fit? No help this time.

23) $5 \div \frac{1}{2} = 10$

24) $3 \div \frac{1}{6} = 18$

25) $6 \div \frac{1}{4} = 24$

26) $2 \div \frac{1}{8} = 16$

27) $7 \div \frac{1}{3} = 21$

28) $4 \div \frac{1}{5} = 20$

29) $8 \div \frac{1}{2} = 16$

30) $5 \div \frac{1}{6} = 30$

31) $9 \div \frac{1}{3} = 27$

32) $2 \div \frac{1}{10} = 20$

33) $6 \div \frac{1}{5} = 30$

34) $4 \div \frac{1}{8} = 32$

Dividing with Unit Fractions: Practice (continued)

Part 5 Check with times

No help this time.

35) $\frac{1}{10} \times 5 = \frac{5}{10}$

36) $\frac{1}{12} \times 4 = \frac{4}{12}$

37) $\frac{1}{15} \times 3 = \frac{3}{15}$

38) $\frac{1}{24} \times 6 = \frac{6}{24}$

39) $\frac{1}{18} \times 3 = \frac{3}{18}$

40) $\frac{1}{18} \times 9 = \frac{9}{18}$

Part 6 Share a piece

41) $\frac{1}{7} \div 5 = \frac{1}{35}$

42) $\frac{1}{10} \div 4 = \frac{1}{40}$

43) $\frac{1}{4} \div 8 = \frac{1}{32}$

44) $\frac{1}{6} \div 8 = \frac{1}{48}$

45) $\frac{1}{9} \div 7 = \frac{1}{63}$

46) $\frac{1}{2} \div 3 = \frac{1}{6}$

Dividing with Unit Fractions: Practice (continued)

Part 6 Share a piece (continued)

47) $\frac{1}{3} \div 9 = \frac{1}{27}$

48) $\frac{1}{6} \div 9 = \frac{1}{54}$

49) $\frac{1}{8} \div 7 = \frac{1}{56}$

50) $\frac{1}{7} \div 6 = \frac{1}{42}$

51) $\frac{1}{7} \div 9 = \frac{1}{63}$

52) $\frac{1}{6} \div 6 = \frac{1}{36}$

53) $\frac{1}{9} \div 8 = \frac{1}{72}$

54) $\frac{1}{10} \div 8 = \frac{1}{80}$

55) $\frac{1}{4} \div 7 = \frac{1}{28}$

56) $\frac{1}{7} \div 3 = \frac{1}{21}$

57) $\frac{1}{8} \div 5 = \frac{1}{40}$

58) $\frac{1}{4} \div 6 = \frac{1}{24}$

Dividing with Unit Fractions: Practice (continued)

Part 6 Share a piece (continued)

59) $\frac{1}{9} \div 5 = \frac{1}{45}$

60) $\frac{1}{2} \div 5 = \frac{1}{10}$

61) $\frac{1}{9} \div 9 = \frac{1}{81}$

62) $\frac{1}{8} \div 6 = \frac{1}{48}$

63) $\frac{1}{7} \div 8 = \frac{1}{56}$

64) $\frac{1}{2} \div 9 = \frac{1}{18}$

Part 7 How many pieces fit?

65) $3 \div \frac{1}{7} = 21$

66) $6 \div \frac{1}{7} = 42$

67) $2 \div \frac{1}{9} = 18$

68) $8 \div \frac{1}{10} = 80$

69) $12 \div \frac{1}{7} = 84$

70) $7 \div \frac{1}{8} = 56$

Dividing with Unit Fractions: Practice (continued)

Part 7 How many pieces fit? (continued)

71) $10 \div \frac{1}{5} = 50$

72) $8 \div \frac{1}{8} = 64$

73) $9 \div \frac{1}{6} = 54$

74) $8 \div \frac{1}{9} = 72$

75) $9 \div \frac{1}{9} = 81$

76) $4 \div \frac{1}{4} = 16$

77) $7 \div \frac{1}{10} = 70$

78) $5 \div \frac{1}{8} = 40$

79) $7 \div \frac{1}{4} = 28$

80) $9 \div \frac{1}{4} = 36$

81) $12 \div \frac{1}{9} = 108$

82) $9 \div \frac{1}{2} = 18$

Dividing with Unit Fractions: Practice (continued)

Part 7 How many pieces fit? (continued)

83) $4 \div \frac{1}{7} = 28$

84) $12 \div \frac{1}{6} = 72$

85) $10 \div \frac{1}{3} = 30$

86) $2 \div \frac{1}{3} = 6$

87) $12 \div \frac{1}{10} = 120$

88) $10 \div \frac{1}{7} = 70$

Part 8 Remember? Problems from earlier packets.

89) $5 \div 7 = \frac{5}{7}$

90) $\frac{5}{6} \times 12 = 10$

91) $\frac{5}{7} - \frac{1}{3} = \frac{8}{21}$

92) $6 \times 0.35 = 2.1$

93) $2.4 + 3.75 = 6.15$

94) $828 \div 36 = 23$

Dividing with Unit Fractions: Practice (continued)

Part 9 Word problems Share it, or how many fit?

95) Half a pizza is shared equally by 3 friends. What part of the whole pizza does each get?

Number sentence 1/2 ÷ 3 = 1/6

Answer 1/6 of the pizza

Draw or show your work

96) How many 1/4 cup scoops are in 5 cups of rice?

Number sentence 5 ÷ 1/4 = 20

Answer 20 scoops

Draw or show your work

97) A 1/3 kg bag of nuts is split into 4 equal bags. How much is in each bag?

Number sentence 1/3 ÷ 4 = 1/12

Answer 1/12 kg

Draw or show your work

98) A 6 m rope is cut into pieces 1/2 m long. How many pieces are there?

Number sentence 6 ÷ 1/2 = 12

Answer 12 pieces

Draw or show your work

99) Challenge: 3 cups of flour, 1/3 cup per cake. Ava made 4 cakes. How many more can she make?

Number sentence 3 ÷ 1/3 = 9, 9 − 4 = 5

Answer 5 cakes

Draw or show your work

Check-In: Dividing with Unit Fractions

Show what you know. No help this time.

Mastery: 6 of 7 (80%)

Part 1 Share a piece

1) 

$\frac{1}{2} \div 4 = \frac{1}{8}$

Share the shaded part among 4.

Part 2 Share a piece No help.

2) $\frac{1}{3} \div 7 = \frac{1}{21}$

3) $\frac{1}{4} \div 4 = \frac{1}{16}$

4) $\frac{1}{5} \div 3 = \frac{1}{15}$

Part 3 How many pieces fit?

5) $6 \div \frac{1}{2} = 12$

6) $3 \div \frac{1}{4} = 12$

7) $4 \div \frac{1}{6} = 24$